

1 Plans

Feel free to add more:

- Use QTT to further compress data (e.g. Q indices in THC)
- Improve constant factors in quantum algorithms
- Reduced norm with new integral kernel (Jiace's) + coherent arithmetic + sparse-encoding
- Compare density fitting with THC (worse scaling but maybe smaller constants/subnormalization)
- More refined recourse estimation for LCC (previously not considered: optimized subnormalization involving ov space, space-time tradeoff, QLSS's dependence on various parameters were not completely right). Perhaps redo one for Femoco.
- Incorporate non-abelian symmetries

2 Reading list

Quantum chemistry:

1. [CCSD equations](#)
2. [Efficient recursive computation of molecular integrals over Cartesian Gaussian functions](#)
3. [Computation of Electron Repulsion Integrals Using the Rys Quadrature Method](#)
4. [An introduction to coupled cluster theory for computational chemists](#)
5. [CC website](#)
6. [CC diagrams](#)
7. [Local correlation methods for periodic systems](#)

Quantum General:

1. [Quantum algorithms: A survey of applications and end-to-end complexities](#)
2. [Hamiltonian simulation in the low energy subspace](#) (mitigate large norm if initial state is in low-energy subspace)

Quantum arithmetic:

1. General: [A Comprehensive Study of Quantum Arithmetic Circuits](#)
2. Multiplication: [Quantum Circuit Design of a T-count Optimized Integer Multiplier](#); [Fast quantum integer multiplication with zero ancillas](#)
3. Division: [Improved quantum circuits for division](#)
4. Adder (see Table 2): [Optimal Toffoli-Depth Quantum Adder](#)
5. Gidney adder: [Halving the cost of quantum addition](#)
6. $2n + O(1)$ Toffoli adder: [A new quantum ripple-carry addition circuit](#)
7. Square root: [Quantum Algorithms and Circuits for Scientific Computing](#); [T-count and Qubit Optimized Quantum Circuit Design of the Non-Restoring Square Root Algorithm](#); [Qrisp Implementation and Resource Analysis of a T-Count-Optimised Non-Restoring Quantum Square-Root Circuit](#)
8. Exponentiation: [Efficient Evaluation of Exponential and Gaussian Functions on a Quantum Computer](#)

Quantum data-loading:

1. [Encoding Electronic Spectra in Quantum Circuits with Linear T Complexity](#) (uniary-iteration QROM, coherent alias sampling)
2. [Trading T gates for dirty qubits in state preparation and unitary synthesis](#)
3. [Black-Box Quantum State Preparation without Arithmetic](#)
4. [Quantum Resources Required to Block-Encode a Matrix of Classical Data](#)
5. [Compilation of Fault-Tolerant Quantum Heuristics for Combinatorial Optimization](#) (Sec II.E, QROM interpolation)
6. [Qubitization of Arbitrary Basis Quantum Chemistry Leveraging Sparsity and Low Rank Factorization](#) (Appendix, cheap uncomputation via measurement and feedback)

First quantization:

1. [Fault-Tolerant Quantum Simulations of Chemistry in First Quantization](#)
2. [Quantum Simulation of Realistic Materials in First Quantization Using Non-local Pseudopotentials](#)
3. [Bounding the costs of quantum simulation of many-body physics in real space](#)
4. [Quantum simulation of electronic structure via quantum fast multipole method](#)

Second quantization:

1. [Fault-tolerant quantum simulation of materials using Bloch orbitals](#)

2. [Even more efficient quantum computations of chemistry through tensor hypercontraction](#)
3. [Qubitization of Arbitrary Basis Quantum Chemistry Leveraging Sparsity and Low Rank Factorization](#) (single factorization)
4. [Quantum computing enhanced computational catalysis](#) (double facorization)
5. [Efficient Simulation of Quantum Chemistry Problems in an Enlarged Basis Set](#) (isometric THC)

Anti-symmetrization:

1. [Quantum Algorithms for Realizing Symmetric, Asymmetric, and Antisymmetric Projectors](#)
2. [Stabilization of Quantum Computations by Symmetrization](#)

3 Quantum algorithm primitives

3.1 Controlled- R_z

The direct synthesis of single-qubit rotation via Clifford + T requires $3 \log_2(1/\epsilon) + O(\log(\log(1/\epsilon)))$ T gates [1]. A controlled single-qubit rotation doubles the number of single-qubit gates (see Theorem 4. of [2] and circuit before it).

3.2 Quantum Shannon decomposition

The quantum Shannon decomposition [2] of a unitary on n qubits ($N = 2^n$) has single qubit gates

$$G_1 = \frac{17}{24}N^2 - \frac{3}{2}N - \frac{1}{3} \quad (1)$$

(reported in Table 5 of [3]), with the rest being Clifford gates (CNOT). We would need to synthesize each operator to error ϵ/G_1 , which results in T count

$$G_1 [3 \log_2(G_1/\epsilon) + O(\log(\log(G_1/\epsilon)))] \quad (2)$$

with direct unitary synthesis. This has a small constant factor, but the multiplicative error term makes it unfavorable.

3.3 Adder

The Gideny-adder [4] has Toffoli count $n-1$ /T-count $4n-4$, and the controlled version doubles it, i.e. Toffoli count $2n-2$ /T-count $8n-8$

3.4 QROM

The unary-iteration QROM [5], for L data, has Toffoli count $L-1$. Note that this number applies even if L is not a power of 2. We can also use the space-time tradeoff QROM of [6], i.e. the Select-Swap network, which takes (see App. B of [7])

$$\left\lceil \frac{L}{\lambda} \right\rceil + b(\lambda - 1) \quad (3)$$

where λ is a power of 2, and requires $b(\lambda - 1) + \lceil \log_2(L/\lambda) \rceil$ **clean** ancillae. The App. C of [7] removes the dependence on b in **uncomputation only**, by using measurement-and-feedforward, which requires

$$\left\lceil \frac{L}{\lambda} \right\rceil + \lambda \quad (4)$$

with $\lambda + \lceil \log_2(L/\lambda) \rceil$ clean ancillae.

3.5 Multiply-controlled phase gate

A multi-controlled phase gate (Appendix D.1 of [6]) is

$$U = \sum_{x=0}^{2^n-1} |x\rangle\langle x| \otimes e^{i2\pi\theta_x Z}, \quad (5)$$

where the angles are specified through QROM. The circuit consists of a forward QRAM computation, a controlled adder acting on a Fourier state ancilla, and QRAM uncomputation.

3.6 State preparation

Consider the approach described in App. D. of [6]. We will apply multiply-controlled phase gates (with single-qubit Cliffords to change Z to Y) in a staircase pattern, with angles specified by b bits, acting on $1, 2, 3, \dots, n$ qubits, each of which requires $1, 2, 4, \dots, 2^{n-1}$ amount of data loading.

We can choose to minimize the Toffoli count as

$$\sum_{i=0}^{n-1} \left(2 \left\lceil \frac{2^i}{\lambda_i} \right\rceil + b(\lambda_i - 1) + \lambda_i \right) \quad (6)$$

(2 comes from two QROM calls) where the the minimizer is

$$\log_2(\lambda_i) = \left\lceil \frac{i - \log_2(b+1)}{2} \right\rceil \quad (7)$$

(Demo). If we were instead trying to minimize the space, then we would use the unary-iteration idea [5], and the total Toffoli count is

$$S_{\text{minSpace}}(n, c) = cn(b-1) + \sum_{i=0}^{n-1} 2(2^i - 1) = cn(b-1) + 2^{n+1} - 2n - 2 \quad (8)$$

with $b+n-1$ ancillas and a b ancilla Fourier state.

3.7 Unitary synthesis

We can construct a unitary using controlled state-preparation unitaries for each column (Sec. 4 of [6]), by constructing the reflection of each state. Specifically, let A_k be the state preparation unitary of column k , i.e.

$$A_k |0\rangle = U |k\rangle = |u_k\rangle. \quad (9)$$

Consider the state

$$|w_k\rangle = \frac{1}{\sqrt{2}}(|1\rangle \otimes |k\rangle - |0\rangle \otimes |u_k\rangle) \quad (10)$$

It is easy to verify that $\langle w_k | w_{k'} \rangle = \delta_{k,k'}$, and thus one can prepare

$$\prod_{k=0}^{M-1} (I - 2 |w_k\rangle \langle w_k|) \quad (11)$$

which acts identically as

$$W = |0\rangle \langle 1| \otimes U + |1\rangle \langle 0| \otimes U^\dagger \quad (12)$$

on states $|w_k\rangle$. Given a state-preparation unitary A_k satisfying $A_k |0\rangle = |u_k\rangle$, one can prepare $|w_k\rangle$ starting from $|0\rangle \otimes |e_1\rangle$ as follows: apply a Hadamard gate to the first qubit, then a sequence of CNOT gates to prepare $(|1\rangle \otimes |k\rangle + |0\rangle \otimes |0\rangle)/\sqrt{2}$, and finally apply A_k controlled on the first qubit with a negative control (i.e., conditioned on $|0\rangle$). Call the circuit A'_k , then the reflection can be constructed from

$$I - 2 |w_k\rangle \langle w_k| = A'_k (I - 2 |0\rangle \langle 0|) A_k^{\dagger} \quad (13)$$

Each query is twice the controlled-state-preparation unitary. We also need the reflection $I - 2 |0\rangle \langle 0|$, where the controlled version has at most $2n - 5$ Toffolis. So with M columns specified, the total cost is

$$U_*(n) = M(2S_*(n, 2) + 2n - 5) \quad (14)$$

with $*$ chosen to minimize Toffoli count or space.

In the paper [8] the unitary synthesis has constant factor improved by $\times 7$.

3.8 Block-encoding by unitary synthesis

Suppose we are given a matrix A of size $N \times M$, and we wish to block-encode $A/\|A\|$ in a unitary. We can achieve this by finding B such that

$$B^\dagger B + \frac{A^\dagger A}{\|A\|^2} = I_M \quad (15)$$

and construct the isometry

$$V = \begin{bmatrix} \frac{A}{\|A\|} \\ B \end{bmatrix} \quad (16)$$

which specifies the first M columns of the unitary. Let the SVD of A be $A = U\Sigma W^\dagger$, where Σ is square and has dimension $\min(N, M)$. Then B can be constructed from

$$B = \sqrt{1 - \frac{\Sigma}{\|A\|^2}} W^\dagger \quad (17)$$

so the total shape of V is $(N + \min(N, M)) \times M$, and we can pad zeros if we demand the number of rows is a power of two, giving a number of qubits $\lceil \log_2(N + \min(N, M)) \rceil$.

3.9 Block-encoding the $|0\rangle \langle 0|$ and $I - |0\rangle \langle 0|$ projector

We can apply a multi-controlled-not, controlled on $|0\rangle$ and targeted on an ancilla starting with $|0\rangle$, and post-select the ancilla with $|1\rangle$ to get a block-encoding of $|0\rangle \langle 0|$ (see circuit). For any projector $|\psi\rangle \langle \psi|$ where the state-preparation unitary $|\psi\rangle = U |0\rangle$ is simple, we can use $U |0\rangle \langle 0| U^\dagger$ for its block-encoding. The block-encoding $I - |0\rangle \langle 0|$ can simply be obtained by post-selecting the ancilla with $|0\rangle$ (see circuit).

3.10 Quantum linear system solver

Given a block-encoding of A/α , let A 's condition number be κ . Most quantum algorithms depend on the joint quantity (see Page 288 of [9])

$$\kappa' = \kappa \frac{\alpha}{\|A\|} = \alpha \|A^{-1}\| = \frac{\alpha}{\sigma_{\min}} \quad (18)$$

and the optimal discrete adiabatic algorithm [10] has dependence

$$O\left(\kappa' \log\left(\frac{1}{\epsilon}\right)\right) = O\left(\kappa \frac{\alpha}{\|A\|} \log\left(\frac{1}{\epsilon}\right)\right). \quad (19)$$

If one uses QSVT instead, one must approximate the bounded polynomial

$$p(x) \approx \frac{1}{\kappa' x} \quad (20)$$

so that $p(\sigma_{\min}/\alpha) \approx 1$, where the degree is $O(\kappa' \log(\kappa'/\epsilon))$ (Corollary 26 of [11]). However, one must also consider the post-selection amplitude associated with

$$p\left(\frac{A}{\alpha}\right) |b\rangle \approx \frac{1}{\kappa'} \alpha A^{-1} |b\rangle = \frac{\sigma_{\min}}{\alpha} \alpha A^{-1} |b\rangle = \sigma_{\min} A^{-1} |b\rangle \quad (21)$$

where the lower bound is associated with the *largest* singular value

$$\sqrt{p_{\text{succ}}} = \|\sigma_{\min} A^{-1} |b\rangle\| \geq \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{1}{\kappa} \quad (22)$$

so using amplitude amplification which has query complexity $O(1/\sqrt{p_{\text{succ}}})$ (see [notes](#) here), the worst-case total complexity for QSVT is

$$O\left(\kappa \kappa' \log\left(\frac{\kappa'}{\epsilon}\right)\right) = O\left(\kappa^2 \frac{\alpha}{\|A\|} \log\left(\frac{\kappa \alpha}{\epsilon \|A\|}\right)\right) \quad (23)$$

Note that for PSD matrices, classical conjugate gradient methods can achieve $O(\sqrt{\kappa})$ dependence. This is due to that $1/x$ has a good polynomial approximation in $[1/\kappa, 1]$, but a bounded polynomial in $[-1, 1]$ must have $O(\kappa)$ degree (see text after Corollary 29 [here](#)). For certain PSD (not all) matrices, quantum algorithms can also achieve $O(\sqrt{\kappa})$ dependence, see [here](#), e.g. when *normalized block-encoding* $(I - \eta A)$ is accessible (not $(I - \eta A)/c$ for some constant c).

4 Cost of LCC THC on FeMoco

The block-encoding encodes a linear combination of 35 operators, with coefficients listed below. The Hamiltonian does not preserve (quasi-)particle number, but one can introduce fictitious modes. For $m = 4$, we introduce states $|\perp_1^o\rangle, |\perp_2^o\rangle, |\perp_3^o\rangle$ and $|\perp_1^v\rangle, |\perp_2^v\rangle, |\perp_3^v\rangle$ that share the same anti-symmetric rules with other fermions.

Coefficient	$k = 1$	$k = 2$	$k = 3$	$k = 4$	General	Diagram
(1) $F_{oo,k \rightarrow k}$	1	2	3	4	k	
(2) $F_{vv,k \rightarrow k}$	1	2	3	4	k	$\ F\ _2$
(3) $F_{ov,k \rightarrow k-1}$		1	1	1	1	$\ F\ _F$
(4) $\langle oo oo\rangle, k \rightarrow k$		$\frac{1}{2}$	$\frac{3}{2}$	3	unknown	$\max_{ij} \zeta_{ij} \ \chi\ _2^4$
(5) $\langle vv vv\rangle, k \rightarrow k$		$\frac{1}{2}$	$\frac{3}{2}$	3	unknown	$\max_{ij} \zeta_{ij} \ \chi\ _2^4$
(6) $\langle ov ov\rangle, k \rightarrow k$	-1	-4	-9	-16	$-k^2$	$(\max_{ij} \zeta_{ij} + \ \zeta\ _2) \ \chi\ _2^4$
(7) $\langle oo ov\rangle, k \rightarrow k-1$		$-\frac{1}{2}$	-1	$-\frac{3}{2}$	$-(k-1)/2$	$(\max_{ij} \zeta_{ij} + \ \zeta\ _2) \ \chi\ _2^4$
(8) $\langle ov vv\rangle, k \rightarrow k-1$		$-\frac{1}{2}$	-1	$-\frac{3}{2}$	$-(k-1)/2$	$(\max_{ij} \zeta_{ij} + \ \zeta\ _2) \ \chi\ _2^4$
(9) $\langle oo ov\rangle, k-1 \rightarrow k$		-2	-9	-24	$-k^2(k-1)/2$	$(\max_{ij} \zeta_{ij} + \ \zeta\ _2) \ \chi\ _2^4$
(10) $\langle ov vv\rangle, k-1 \rightarrow k$		-2	-9	-24	$-k^2(k-1)/2$	$(\max_{ij} \zeta_{ij} + \ \zeta\ _2) \ \chi\ _2^4$
(11) $\langle oo vv\rangle, k \rightarrow k-2$			$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\ \zeta\ _F \ \chi\ _2^4$

Table 1: Prefactors for operators before anti-symmetrization, and the norm one should consider.

The types of operators we need to compile include:

- Row (1) to (3): $\sum_{ij} F_{ij} |i\rangle \langle j|$, compile via unitary synthesis.
- Row (4) to (11): $\sum_{ij} \chi_{ij} |i\rangle \langle j|$, compile via unitary synthesis.
- Row (4) to (11): $\sum_{ij} \zeta_{ij} |ij\rangle \langle ij|$ (this comes from anti-symmetrized integral), compile via two QROMs and rotation.
- Row (4) to (10): $\sum_{ij} \zeta_{ij} |i\rangle \langle j|$, compile via unitary synthesis.
- Row (11): An outer product of $|\zeta\rangle = \sum_{ij} \zeta_{ij} |ij\rangle$ and a trivial state (involving $|\perp^o\rangle$ and $|\perp^v\rangle$). This can be achieved using a block encoding of $|0\rangle \langle 0|$ and a state preparation unitary of $|\zeta\rangle$.

Using the [THC data](#) of FeMoco from [12], and for the suggested THC rank = 450 (See Sec IV.A of [12]), we obtain the following values

- $\|\chi\|_2 = 1.7599927190414546$
- $\max_{ij} |\zeta_{ij}| = 2.472872105787373$
- $\|\zeta\|_2 = 24.86573725010864$
- $\|\zeta\|_F = 41.515473867785225$

We assume F do not contribute to the final cost much and leave it aside for now. We multiply the absolute values of the prefactors of operators in Table 1 and obtain the total subnormalization

$$10(\max_{ij} |\zeta_{ij}| \|\chi\|_2^4 + 106(\max_{ij} |\zeta_{ij}| + \|\zeta\|_2) \|\chi\|_2^4 + \frac{1}{2} \|\zeta\|_F \|\chi\|_2^4 = 28241.624804503917 \quad (24)$$

Now for the synthesis cost, we set $\epsilon = 10^{-6}$ and obtain the following

- $\sum_{ij} \chi_{ij} |i\rangle \langle j|$: Substitute $N = 512$ (first power of two > 450) and $K = 76$ to Eq. (14), we get 936,049.
- $\sum_{ij} \zeta_{ij} |ij\rangle \langle ij|$: Substitute $L = 450^2$ to Eq. (??) and $\times 2$, we get 1,619,992.
- $\sum_{ij} \zeta_{ij} |i\rangle \langle j|$: Substitute $N = 512$ and $K = 450$ to Eq. (14), we get 5,542,394
- $|\zeta\rangle = \sum_{ij} \zeta_{ij} |ij\rangle$: Substitute $N = 262144$ (first power of two $> 450^2$) we get 2,100,852

Each of the operators ζ above can be prepared one time and swapped to the corresponding positions at the level of using a linear combination of block-encodings (for some basic info on the linear combination, see Lemma 1.7 of [notes](#)). χ can be prepared 8 times (on the first two particle-hole pairs) and chosen to be turn on/off. So in total we get T count 16,751,630. Multiplying the total subnormalization, we get

$$4.73 \times 10^{11} \tag{25}$$

T gates.

5 LCC formalism

The complete effective Hamiltonian operator is written as sum of local operators O_i (local means apply on $O(1)$ first-quantized quasi-particles), followed by a global antisymmetrization:

$$T' = \text{Antisym} \left(\sum_{i=1}^{11} c_i \hat{O}_i T \right)$$

where each $\hat{O}_i$ is at most 2-local without summing over permutations, and

$$\text{Antisym} \left(T^{(k)} \right)_{i_1 \dots i_k a_1 \dots a_k} = \frac{1}{(k!)^2} \sum_{\sigma \tau} (-1)^{s(\sigma)+s(\tau)} P(\sigma) P(\tau) T_{i_1 \dots i_k a_1 \dots a_k}^{(k)}$$

has exactly $\|\text{Antisym}\|_2 = 1$ (it functions as a projection operator in the antisymmetrized space).

Without loss of generality, one can choose that all operators act on the first 2 quasiparticles (as all choices are equivalent after antisymmetrization). Thus each $\hat{O}$ has

$$\hat{O} = \hat{O}_{1,2} \otimes I_3 \otimes \dots \otimes I_k,$$

whose norm is completely determined by $\hat{O}_{1,2}$ and its coefficients. Since $\hat{O}_{1,2}$ does not depend on k , the k -dependence only comes from the coefficients.

Here's the list of 11 operators and their coefficients up to $k = 4$ (LCCSDTQ). The coefficients are fitted numerically by (1) evaluating $\langle k|[H, T]|0\rangle$ in the FCI space (verified again pyscf CCSD), (2) linear fit the coefficients of each diagram to match $\langle k|[H, T]|0\rangle$ result from FCI reference). The general expression is guessed from the data points of $k \leq 4$.

coefficient	$k = 1$	$k = 2$	$k = 3$	$k = 4$	general	Expr	leading ζ -related norm
$F_{oo}, k \rightarrow k$	1	2	3	4	k	$F_{ij} T_{ia\dots} \rightarrow T_{ja\dots}$	
$F_{vv}, k \rightarrow k$	1	2	3	4	k	$F_{ab} T_{ia\dots} \rightarrow T_{ib\dots}$	
$F_{ov}, k \rightarrow k - 1$		1	1	1	1	$F_{ia} T_{ia\dots} \rightarrow T_{\dots}$	
$\langle oo oo\rangle, k \rightarrow k$		1/2	3/2	3	$k(k-1)/4$	$\langle kl ij\rangle T_{iajb\dots} \rightarrow T_{kalb\dots}$	$\max_{ij} \zeta_{ij} $
$\langle vv vv\rangle, k \rightarrow k$		1/2	3/2	3	$k(k-1)/4$	$\langle cd ab\rangle T_{iajb\dots} \rightarrow T_{icjd\dots}$	$\max_{ij} \zeta_{ij} $
$\langle ov ov\rangle, k \rightarrow k$	-1	-4	-9	-16	$-k^2$	$\langle jc kb\rangle T_{iajb\dots} \rightarrow T_{iakc\dots}$	$\ \zeta\ _2$
$\langle oo ov\rangle, k \rightarrow k - 1$		-1/2	-1	-3/2	$-(k-1)/2$	$\langle ij kb\rangle T_{iajb\dots} \rightarrow T_{ka\dots}$	$\ \zeta\ _2$
$\langle ov vv\rangle, k \rightarrow k - 1$		-1/2	-1	-3/2	$-(k-1)/2$	$\langle jc ab\rangle T_{iajb\dots} \rightarrow T_{ic\dots}$	$\ \zeta\ _2$
$\langle oo ov\rangle, k - 1 \rightarrow k$		-2	-9	-24	$-k^2(k-1)/2$	$\langle jk ib\rangle T_{ia\dots} \rightarrow T_{jkab\dots}$	$\ \zeta\ _2$
$\langle ov vv\rangle, k - 1 \rightarrow k$		-2	-9	-24	$-k^2(k-1)/2$	$\langle ja bc\rangle T_{ia\dots} \rightarrow T_{ijbc\dots}$	$\ \zeta\ _2$
$\langle oo vv\rangle, k \rightarrow k - 2$			1/4	1/4	1/4	$\langle ij ab\rangle T_{iajb\dots} \rightarrow T_{\dots}$	$\ \zeta\ _F$

If one redefine $T'_k = \frac{1}{k!} T_k$, the operator coefficients change as $O'_{k \rightarrow k'} = \frac{k!}{k'!} O_{k \rightarrow k'}$. The new coefficients are

coefficient	$k = 1$	$k = 2$	$k = 3$	$k = 4$	general	Expr	leading ζ -related norm
$F_{oo}, k \rightarrow k$	1	2	3	4	k	$F_{ij} T_{ia\dots} \rightarrow T_{ja\dots}$	
$F_{vv}, k \rightarrow k$	1	2	3	4	k	$F_{ab} T_{ia\dots} \rightarrow T_{ib\dots}$	
$F_{ov}, k \rightarrow k - 1$		2	3	4	k	$F_{ia} T_{ia\dots} \rightarrow T_{\dots}$	
$\langle oo oo\rangle, k \rightarrow k$		1/2	3/2	3	$k(k-1)/4$	$\langle kl ij\rangle T_{iajb\dots} \rightarrow T_{kalb\dots}$	$\max_{ij} \zeta_{ij} $
$\langle vv vv\rangle, k \rightarrow k$		1/2	3/2	3	$k(k-1)/4$	$\langle cd ab\rangle T_{iajb\dots} \rightarrow T_{icjd\dots}$	$\max_{ij} \zeta_{ij} $
$\langle ov ov\rangle, k \rightarrow k$	-1	-4	-9	-16	$-k^2$	$\langle jc kb\rangle T_{iajb\dots} \rightarrow T_{iakc\dots}$	$\ \zeta\ _2$
$\langle oo ov\rangle, k \rightarrow k - 1$		-1	-3	-6	$-k(k-1)/2$	$\langle ij kb\rangle T_{iajb\dots} \rightarrow T_{ka\dots}$	$\ \zeta\ _2$
$\langle ov vv\rangle, k \rightarrow k - 1$		-1	-3	-6	$-k(k-1)/2$	$\langle jc ab\rangle T_{iajb\dots} \rightarrow T_{ic\dots}$	$\ \zeta\ _2$
$\langle oo ov\rangle, k - 1 \rightarrow k$		-1	-3	-6	$-k(k-1)/2$	$\langle jk ib\rangle T_{ia\dots} \rightarrow T_{jkab\dots}$	$\ \zeta\ _2$
$\langle ov vv\rangle, k - 1 \rightarrow k$		-1	-3	-6	$-k(k-1)/2$	$\langle ja bc\rangle T_{ia\dots} \rightarrow T_{ijbc\dots}$	$\ \zeta\ _2$
$\langle oo vv\rangle, k \rightarrow k - 2$			3/2	3	$k(k-1)/4$	$\langle ij ab\rangle T_{iajb\dots} \rightarrow T_{\dots}$	$\ \zeta\ _F$

6 Estimation using planewave basis

Setup: $L = 10A \approx 19B$ (A for angstrom, B for Bohr radius), $E_{\text{cutoff}} = 100E_h$.

Final cost = Data loading $\times 20 \times$ Projector $\times$ Coulomb norm $\times$ polynomial order $\times$ Phase estimation

1. Data loading: $\Delta G = \frac{2\pi}{L} \approx 0.33B^{-1}$, $G_{\text{max}} = \sqrt{2E_{\text{cutoff}}} \approx 14B^{-1}$, $\Delta L = \frac{2\pi}{G_{\text{max}}} = 0.45B$, $\frac{G_{\text{max}}}{\Delta G} \approx 42$, $N = \frac{4\pi}{3} \left(\frac{G_{\text{max}}}{\Delta G} \right)^3 \approx \mathbf{3.2 \times 10^5}$,
2. Projector cost:
 - (a) conservative: $\epsilon = 10^{-4} \rightarrow d \sim \log(1/\epsilon) \sim 10$, $\kappa = \frac{E_{\text{cutoff}}}{0.1} = 10^3$, cost: $10\kappa d \sim \mathbf{10^5}$
 - (b) aggressive + modify high-energy spectrum + large gap: $\epsilon = 10^{-2} \rightarrow d \sim \log(1/\epsilon) \sim 5$, $\kappa = \frac{0.2 \times E_{\text{cutoff}}}{0.5} = 40$, cost: $5\kappa d \sim \mathbf{10^3}$
3. Coulomb operator norm (see derivation below): $\|V\|_2 \approx 2\pi \left(\frac{L}{2\Delta L} \right)^2 \times \frac{1}{\Delta L} \approx 6 \times 10^3$, normalization factor of LCC 2-body Hamiltonian $\sim 100\|V\|_2 \sim \mathbf{6 \times 10^5}$
4. Phase estimation cost:
 - (a) conservative, $\epsilon = 10^{-3}$, $1/\epsilon \sim \mathbf{10^3}$
 - (b) aggressive, $\epsilon = 0.05$, $1/\epsilon \sim \mathbf{20}$

Final cost:

- conservative: $\mathbf{4 \times 10^{20}}$ $\times$ polynomial order
- aggressive: $\mathbf{10^{17}}$ $\times$ polynomial order

Derivation of $\|V\|_2$: using a rectangle-shaped delta function: $f(x) = \frac{1}{\sqrt{\Delta x}} h(|x - x_0| - \Delta x/2)$, such that $|x\rangle$ is orthonormal basis. So in this case $\langle xx' | \frac{1}{r} | xx' \rangle = \frac{1}{|x - x'|}$. Let the 2-norm of Coulomb on the grid be $\text{norm}(\Delta L, L)$. Clearly $\text{norm}(\Delta L, L) = \frac{1}{\Delta L} \text{norm}(1, \frac{L}{\Delta L})$. To compute $\text{norm}(1, \frac{L}{\Delta L})$, the corresponding matrix is basically translational invariant, and the largest eigenvector is almost uniform. So

$$\text{norm}(1, 2R) \approx \sum_{|r| \leq R} \frac{1}{r} \approx \int_0^R \frac{1}{r} dV = 2\pi R^2$$

Thus

$$\text{norm}(\Delta L, L) \approx \frac{1}{\Delta L} \frac{\pi}{2} \left(\frac{L}{\Delta L} \right)^2$$

7 THC+PBC+BSE

Consider $N_k = 6 \times 6 \times 6 = 216$ supercell/k-points, each unit cell has 2 atoms, so totally $432 \approx 500$ atoms. For Gaussian-type basis (consider dz-like basis here), there are $N_{AO} \sim 5000$ orbitals. Assume the THC rank is $N_{\text{THC}} \sim 10N_{AO} \sim 50000$. For $m = 3$, the baseline number of qubits is $n = 2m \log N_{AO} \approx \mathbf{80}$.

Data loading: All THC tensors are translation invariant. Considering the largest tensor ζ , the number of independent elements of ζ is $N_\zeta \sim \frac{N_{\text{THC}}^2}{N_k} \sim 10^7$. We consider directly loading all classical data without any coherent computing. With the trading approach, one can achieve λ -times reduction with $\sim 32\lambda$ ancilla qubits. (32 for the single precision) Let $\lambda = 10$ (corresponding to **320 ancilla qubits**), the total data loading cost is then $\sim 20N_\zeta/\lambda \sim \mathbf{2 \times 10^7}$ (20 is some synthesis constant).

Sub-normalization: The max Fock eigenvalue is < 5 Hartree in Gaussian-type basis. The $(ov|ov)$ type integrals (dipole-dipole) has norm < 10 . With some THC fitting of $(ov|ov)$ specifically, we can probably achieve total Coulomb norm ~ 10 . Assume BSE has ~ 10 terms, the total sub-normalization factor is $\sim 10 \times 10 \sim \mathbf{10^2}$.

Phase estimation: $\|H\|_2$ is already considered in the sub-normalization, so the phase estimation cost is simply $1/\epsilon$. Let $\epsilon \sim 10^{-3}$, the phase estimation cost is $\sim \mathbf{10^3}$.

Total: gate count is $2 \times 10^7 \times 10^2 \times 10^3 = \mathbf{2 \times 10^{12}}$, number of qubits is $\sim 80 + 320 + \text{extra} \sim \mathbf{500}$.

8 THC numerical optimization

Diamond in dzvp basis ($N_o = 4, N_v = 22$). Error and norm are defined by

$$\begin{aligned}\text{ERI error} &= \frac{\|\Delta(ov|vo)\|_F}{\|(ov|vo)\|_F}, \\ \text{norm} &= \|X_{iI}\|_2^2 \|X_{aI}\|_2^2 \|W_{IJ}\|_2, \\ \text{other norm} &= \|X_{iI}\|_2^2 \|X_{aI}\|_2^2 \max_{IJ} W_{IJ},\end{aligned}$$

The used techniques include (1) ISDF fit for $(ov|vo)$ and $(ov|ov)$, (2) Regularize the linear system solver in ISDF, (3) full nonlinear optimization. The THC rank is $N_{\text{THC}} = cN_{AO}$.

Comparison of optimization strategies for $2 \times 2 \times 2$ diamond with fixed $c = 10$.

Strategy	None	(1)	(1)+(2)	(1)+(2)+(3)
ERI error	7.3×10^{-3}	1.6×10^{-3}	1.8×10^{-3}	2.8×10^{-3}
norm	24000	14000	1000	27

All results below use (1)+(2)+(3), with loss $L = L_{\text{ERI error}} + \alpha L_{\text{norm}}/N_k$.

Comparison of different c for $2 \times 2 \times 2$ diamond, with fixed $\alpha = 0.1$.

CCSD energy error = replace $(ov|vo)$ part with THC, other parts keep unchanged.

c_{IP}	4	5	6	8	10
ERI error	1.3×10^{-2}	7.6×10^{-3}	5.7×10^{-3}	5.1×10^{-3}	4.0×10^{-3}
CCSD energy error	3.0×10^{-5}	2.3×10^{-5}	7.39×10^{-6}	3.64×10^{-6}	1.3×10^{-6}
norm	26	29	31	31	31

Comparison of different α for $2 \times 2 \times 2$ diamond, with fixed $c = 10$

α	0.1	0.2
ERI error	2.8×10^{-3}	1.0×10^{-2}
norm	27	21

Results of different k points

$c = 10, \alpha = 0.1$

k points	$2 \times 2 \times 2$	$3 \times 3 \times 3$	$4 \times 4 \times 4$	$5 \times 5 \times 5$
ERI error	3.0×10^{-3}	3.1×10^{-3}	3.0×10^{-3}	2.8×10^{-3}
norm	32	74	122	166

$c = 5, \alpha = 0.1$

k points	$1 \times 1 \times 1$	$2 \times 2 \times 2$	$3 \times 3 \times 3$	$4 \times 4 \times 4$	$5 \times 5 \times 5$	$6 \times 6 \times 6$
ERI error	2.5×10^{-3}	6.6×10^{-3}	5.7×10^{-3}	7.5×10^{-3}	7.7×10^{-3}	6.9×10^{-3}
norm	4.4	29	72	108	154	233
other norm	3.1	21.3	8.8	11.0	12.6	

$c = 5, \alpha = 0.02N_k^{2/3}$

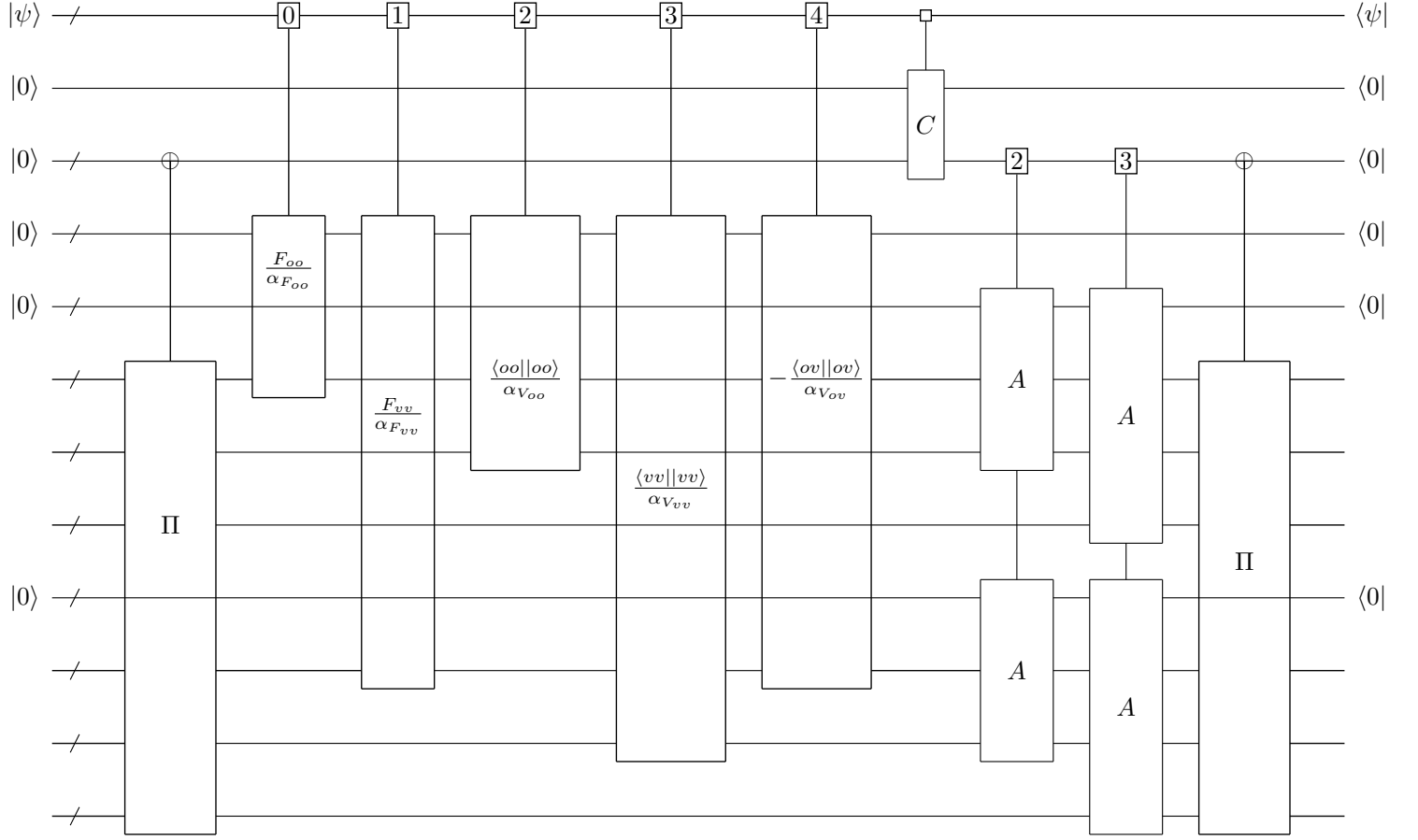
k points	$1 \times 1 \times 1$	$2 \times 2 \times 2$	$3 \times 3 \times 3$	$4 \times 4 \times 4$	$5 \times 5 \times 5$	$6 \times 6 \times 6$
ERI error	1.0×10^{-5}	5.5×10^{-3}	1.2×10^{-2}	1.8×10^{-2}	1.8×10^{-2}	2.0×10^{-2}
norm	4.5	31	47	63	75	79
other norm	2.8	18.7	12.1	15.8	11.1	

True 2-norms of $(ov|vo)$

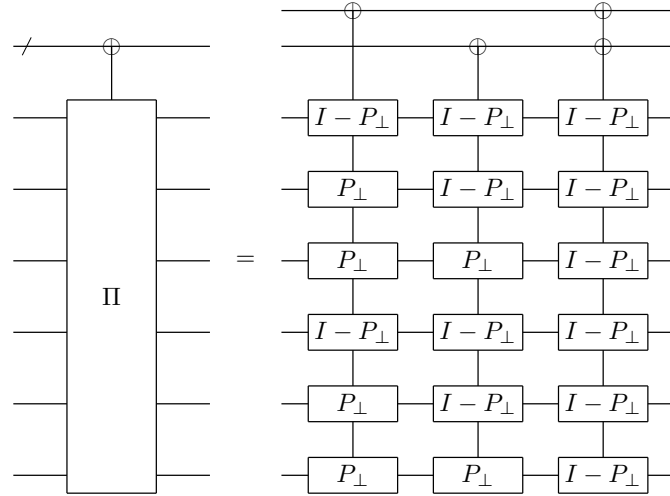
$1 \times 1 \times 1$	$2 \times 2 \times 2$	$3 \times 3 \times 3$	$4 \times 4 \times 4$
0.4726	0.4525	0.4525	0.4726

9 Block-encoding schemes for BSE and LCC

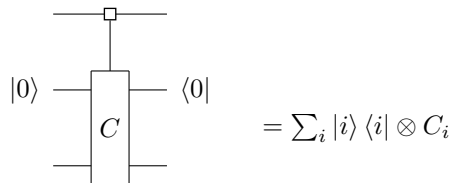
The full block-encoding circuit for $m = 3$ is as follows



The first register is an ancilla used to sum over block-encoding of each type of operator. The state $|\psi\rangle$ is to be specified later. A is the antisymmetrizer which can be block-encoded by the scheme in [13]. The component labeled Π is the particle number counter, assigning the particle count to the ancilla targeted by $\oplus$. For $m = 3$ we have



where $P_{\perp}$ and $I - P_{\perp}$ indicating controlled on “no particle” and “with particles” spaces, and the operators count particle numbers to be 1, 2, 3 respectively from left to right. C is a multiplex-controlled operator that encodes the weights of different particle sectors for each operator type (see Sec. 5):



where for $m = 3$ we have C_i defined as follows.

$$\begin{aligned}
C_0 &= \frac{1}{3} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} & C_1 &= \frac{1}{3} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \\
C_2 &= \frac{2}{3} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{3}{2} \end{pmatrix} & C_3 &= \frac{2}{3} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{3}{2} \end{pmatrix} \\
C_4 &= \frac{1}{9} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 9 \end{pmatrix}
\end{aligned} \tag{26}$$

Then we have the state $|\psi\rangle$ to produce the correct weight for each operator, with amplitude being subnormalization of each operator times the maximal coefficient among particle sectors. For $m = 3$,

$$|\psi\rangle = \frac{1}{\sqrt{\alpha}} (\sqrt{3\alpha_{F_{oo}}} |0\rangle + \sqrt{3\alpha_{F_{vv}}} |1\rangle + \sqrt{\frac{3}{2}\alpha_{V_{oo}}} |2\rangle + \sqrt{\frac{3}{2}\alpha_{V_{vv}}} |3\rangle + \sqrt{9\alpha_{V_{ov}}} |4\rangle) \tag{27}$$

which gives the total subnormalization

$$\alpha = 3\alpha_{F_{oo}} + 3\alpha_{F_{vv}} + \frac{3}{2}\alpha_{V_{oo}} + \frac{3}{2}\alpha_{V_{vv}} + 9\alpha_{V_{ov}} \tag{28}$$

10 BSE THC

Notations:

1. N_{AO} : number of AOs per unit cell.
2. N_k : number of k-points/supercell size
3. $N_{IP} = c_{IP}N_{AO}$: number of interpolating points/THC rank (not including N_k). c_{IP} is a constant (e.g. 8).
4. N_o, N_v : number of occupied and virtual orbitals **per unit cell**.
5. b : precision used for arithmetic, default $b = 32$.

We work with the THC form

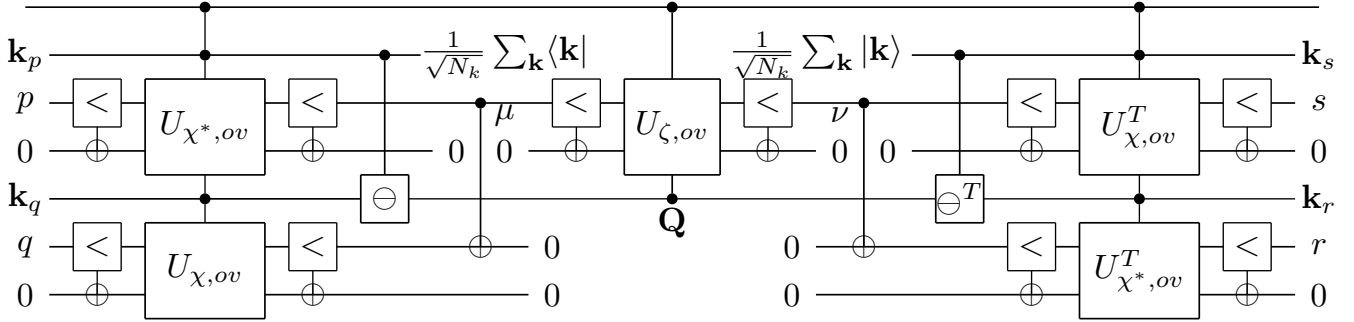
$$(p\mathbf{k}_p, q\mathbf{k}_q | r\mathbf{k}_r, s\mathbf{k}_s) = \sum_{\mu, \nu} \chi_{p, \mathbf{k}_p, \mu}^* \chi_{q, \mathbf{k}_q, \mu} \zeta_{\mathbf{Q}, \mu, \nu} \chi_{r, \mathbf{k}_r, \nu}^* \chi_{s, \mathbf{k}_s, \nu} \quad (29)$$

where $\mathbf{Q} = \mathbf{k}_p - \mathbf{k}_q + \mathbf{B}_n = \mathbf{k}_s - \mathbf{k}_r + \mathbf{B}_m$, with $\mathbf{B}_n$ and $\mathbf{B}_m$ being integer multiples of lattice vectors such that $\mathbf{Q}$ is shifted to the first Brillouin zone. This is equivalent to modulo subtraction along each dimension, and we use the notation $\mathbf{Q} = \mathbf{k}_p \ominus \mathbf{k}_q = \mathbf{k}_s \ominus \mathbf{k}_r$. Here, the dimension of p is N_{AO} , the dimension of $\mathbf{k}_p$ is N_k , and the rank of μ is N_{IP} .

10.1 Controlled-block-encoding of Coulomb terms

10.1.1 The $(ov|vo)$ diagram

We first consider block-encoding the $(ov|vo)$ diagram. The circuit looks like the following



where the components are:

1. $U_{\chi, ov}$: the unitary that block-encode the χ tensor.
2. $U_{\zeta, ov}$: the unitary that block-encode the ζ tensor.
3. Dot and $\oplus$: together they represent CNOTs; a single dot indicating multiplex control on the targeting unitaries (i.e. $\sum_x |x\rangle\langle x| \otimes U_x$; a single $\oplus$ indicating output of a certain operation).
4. Operator with $\ominus$ label: modulo subtractor (the transpose is modulo adder). For example, the first subtractor computes $\mathbf{k}_p \ominus \mathbf{k}_q$ and writes it in an ancilla register.
5. Operator with $<$ label: the comparator, which is the mapping: $\sum_k |k\rangle |0\rangle \rightarrow \sum_{k \leq k'} |k\rangle |0\rangle + \sum_{k > k'} |k\rangle |1\rangle$, for some pre-defined k' . This is to construct the projector of the block-encoding.

Controlled-block-encoding of χ We apply block-encoding by unitary synthesis (Sec. 3.8) to the four χ matrices. Let

$$n_{\chi, ko} = \min(\lceil \log_2(N_k) \rceil + \lceil \log_2(N_o) \rceil + 1, \lceil \log_2(2^{\lceil \log_2(N_k) \rceil + \lceil \log_2(N_o) \rceil} + N_{IP}) \rceil) \quad (30)$$

then insert into Eq. (8) and Eq. (14), for one in the the occupied part, the gate count is $N_{IP}(4n_{\chi, ko}(b-1) + 2^{n_{\chi, ko}+1} - n_{\chi, ko} - 1)$. For the virtual part, similarly define $n_{\chi, kv}$ as above with N_o replaced by N_v . So the total Toffoli count for four block encodings is

$$2N_{IP}(4n_{\chi, ko}(b-1) + 2^{n_{\chi, ko}+1} - n_{\chi, ko} + 4n_{\chi, kv}(b-1) + 2^{n_{\chi, kv}+1} - n_{\chi, kv} - 2) \quad (31)$$

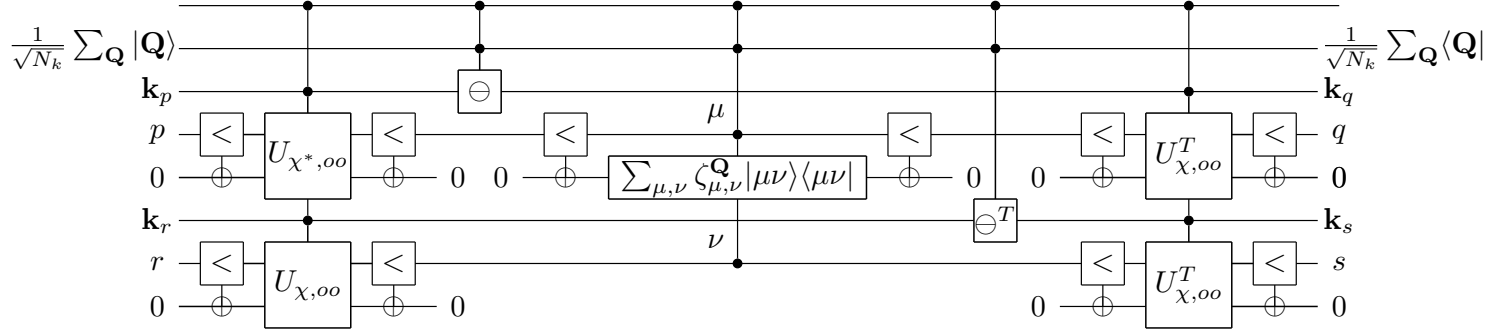
Controlled-block-encoding of ζ This tensor is block-diagonal w.r.t. $\mathbf{Q}$. The implementation only changes QROM, where we take the address to be $|\mathbf{Q}\rangle |\mu\rangle |\nu\rangle$, while the state only acts on μ register. We define $n_\zeta = \lceil \log_2(2N_{IP}) \rceil$. The Toffoli count is

$$N_k N_{IP}(4n_\zeta(b-1) + 2^{n_\zeta+1} - n_\zeta - 1) \quad (32)$$

Other costs The modulo subtractor/adder has Toffoli count $4(\lceil \log_2(N_k) \rceil - 1)$. The CNOTs are ignored. The comparator has linear cost with is $2(\lceil \log_2(N_k) \rceil + \lceil \log_2(N_o) \rceil) + 2(\lceil \log_2(N_k) \rceil + \lceil \log_2(N_v) \rceil) + 6\lceil \log_2(N_{IP}) \rceil$.

10.1.2 The $(oo|oo)$ diagram

The circuit looks like the following



Here slightly different then the above, we assume p register have dimension $\lceil \log_2(N_{IP}) \rceil$ for convenience.

Controlled-block-encoding of χ We define

$$n_{\chi,o} = \lceil \log_2(N_o) \rceil + 1. \quad (33)$$

Similar to before, the Toffoli count for all four block-encoding is

$$4N_k N_{IP} (4n_{\chi,o}(b-1) + 2^{n_{\chi,o}+1} - n_{\chi,o} - 1) \quad (34)$$

Controlled-block-encoding of ζ The ζ matrix is diagonal in this orientation. This is equivalent to the multi-controlled phase gate, changed to Y basis with single-qubit Clifford, and applied on an ancilla with initial state $|0\rangle$ and post-selected with $\langle 0|$. Using Sec. 3.5, we conclude the complexity is

$$N_k N_{IP}^2 + 2b - 2 \quad (35)$$

Other costs The modulo subtractor/adder has Toffoli count $4(\lceil \log_2(N_k) \rceil - 1)$. The comparator has linear cost with is $8(\lceil \log_2(N_k) \rceil + \lceil \log_2(N_{IP}) \rceil) + 2\lceil \log_2(N_{IP}) \rceil$.

10.1.3 The other diagrams

The $(vv|vv)$ diagram has the same structure as $(oo|oo)$, except for changing the occupied dimension to virtual. The $(oo|vv)$ block also has the same diagram as the two mentioned above, except the virtual part is flipped.

10.2 Sector-dependent anti-symmetrizer

We aim to design a special anti-symmetrizer that anti-symmetrizes spaces with different particle numbers separately. Spaces with different particle numbers are labeled by last few registers in state $|\perp\rangle$. For example, if we at most have 3 particles (excitation rank 3), then we use $|i\rangle|\perp\rangle|\perp\rangle$ to represent the 1-particle space and $|i\rangle|j\rangle|\perp\rangle$ for the 2-particle space. We design operators that only anti-symmetrize the first few sites that are not in state $|\perp\rangle$. Define $P_\perp = |\perp\rangle\langle\perp|$ and $Q_\perp = I - P_\perp$. Let the standard anti-symmetrizer denoted by

$$S_m = \frac{1}{m!} \sum_{\pi \in S_m} \text{sgn}(\pi) \hat{\pi} \quad (36)$$

where $\hat{\pi}$ is a permutation operator. The operator we want has form

$$A = A_1 \oplus A_2 \oplus \dots \oplus A_m \quad (37)$$

where

$$A_i = (Q_\perp S_i Q_\perp \otimes P_\perp)_o \otimes (Q_\perp S_i Q_\perp \otimes P_\perp)_v \quad (38)$$

which acts on $2m$ registers, with the first half being the occupied spaces.

10.3 Qubit count

1. System registers: For m quasi-particles, the system requires at least

$$\begin{aligned} & m * \max(\lceil \log_2(N_k) \rceil + \lceil \log_2(N_o) \rceil, \lceil \log_2(N_{IP}) \rceil) \\ & + m * \max(\lceil \log_2(N_k) \rceil + \lceil \log_2(N_v) \rceil, \lceil \log_2(N_{IP}) \rceil) \end{aligned} \quad (39)$$

number of qubits. We also add $2m$ ancillas to easily label $|\perp\rangle$ (i.e., an indicator qubit with 1 labeling the empty state).

2. Block-encoding ancillas:

- 5 from comparators
- 5 from all unitary synthesis.
- $\lceil \log_2(N_k) \rceil$ from $|\mathbf{Q}\rangle$ register.
- $\lceil \log_2(5m - 2) \rceil$ from the sum of diagrams/different number of particles, and one additional to convert multi-control to a single-control on the ancilla.
- $\lceil \log_2(m!) \rceil$ from antisymmetrier.

3. Predefined ancillas: there is one predefined ancilla of b bits, which holds the Fourier state.

4. Reusable ancillas:

- QROM: each address bit needs one ancilla, and plus a register of size b to hold numerical values. The peak usage is thus

$$\lceil \log_2(N_k) \rceil + 2\lceil \log_2(N_{IP}) \rceil + b \quad (40)$$

- Unitary synthesis needs 2 ancillas.
- A register of size $\lceil \log_2(N_k) \rceil$ to hold value $\mathbf{Q}$ for the $(ov|vo)$ diagram.

The total qubit count is the sum of the system register, block-encoding ancillas, predefined ancillas, and the maximum among reusable ancillas.

10.4 Example

Suppose we take diamond with cc-pVDZ basis, and take ISDF/THC rank constant $c_{IP} = 8$, then we have

- $N_k = 6 * 6 * 6 = 216$
- $N_{AO} = 26$
- $N_o = 4$
- $N_v = N_{AO} - N_o = 22$
- $N_{IP} = 8 * N_{AO} = 208$
- $b = 32$

and we assume

- Subnormalization = 10^2
- Phase estimation iteration = 10^3

Substituting into the above equations, we run the following code to a total Toffoli count

```
N_k = 6*6*6
N_AO = 26
N_o = 4
N_v = N_AO - N_o
N_IP = 8 * N_AO
b = 32
```

```
def get_cost(N_k, N_o, N_v, N_IP, b):
    a = np.ceil(np.log2(N_k))
    b = np.ceil(np.log2(N_o))
```

```

n_chi_ko = min(
    a + b + 1,
    np.ceil(np.log2(2 ** (a + b) + N_IP))
)

a = np.ceil(np.log2(N_k))
b = np.ceil(np.log2(N_v))

n_chi_kv = min(
    a + b + 1,
    np.ceil(np.log2(2 ** (a + b) + N_IP))
)

n_zeta = np.ceil(np.log2(N_IP)) + 1
n_chi_o = np.ceil(np.log2(N_o)) + 1
n_chi_v = np.ceil(np.log2(N_v)) + 1

ov_vo_chi = 2 * N_IP * (4 * n_chi_ko * (b - 1) + 2 ** (n_chi_ko + 1) - n_chi_ko + 4 * n_chi_kv * (b - 1) + 2 * n_chi_o)
ov_vo_zeta = N_k * N_IP * (4 * n_zeta * (b - 1) + 2 ** (n_zeta + 1) - n_zeta - 1)

oo_vv_chi = 2 * N_k * N_IP * (4 * n_chi_o * (b - 1) + 2 ** (n_chi_o + 1) - n_chi_o - 1 + 4 * n_chi_v * (b - 1) + 2 * n_chi_ko)
oo_vv_zeta = N_k * N_IP**2 + 2 * b - 2

oo_oo_chi = 4 * N_k * N_IP * (4 * n_chi_o * (b - 1) + 2 ** (n_chi_o + 1) - n_chi_o - 1)
oo_oo_zeta = N_k * N_IP**2 + 2 * b - 2

vv_vv_chi = 4 * N_k * N_IP * (4 * n_chi_v * (b - 1) + 2 ** (n_chi_v + 1) - n_chi_v - 1)
vv_vv_zeta = N_k * N_IP**2 + 2 * b - 2

cost = (ov_vo_chi +
        ov_vo_zeta +
        oo_vv_chi +
        oo_vv_zeta +
        oo_oo_chi +
        oo_oo_zeta +
        vv_vv_chi +
        vv_vv_zeta) * 10**5

return cost

print(f"{get_cost(N_k, N_o, N_v, N_IP, b):e}")

where we get  $1.702226 \times 10^{13}$ , and run the following code to get qubit count

m = 3
N_k = 6*6*6
N_A0 = 26
N_o = 4
N_v = N_A0 - N_o
N_IP = 8 * N_A0
b = 32

sys = m * max(np.ceil(np.log2(N_k)) + np.ceil(np.log2(N_o)), np.ceil(np.log2(N_IP))) + \
    m * max(np.ceil(np.log2(N_k)) + np.ceil(np.log2(N_v)), np.ceil(np.log2(N_IP))) + \
    2 * m

be = 10 + np.ceil(np.log2(N_k)) + np.ceil(np.log2(5*m-2)) + np.ceil(np.log2(math.factorial(m)))

fourier = b

qrom = np.ceil(np.log2(N_k)) + 2*np.ceil(np.log2(N_IP)) + b

print(sys + be + fourier + qrom)

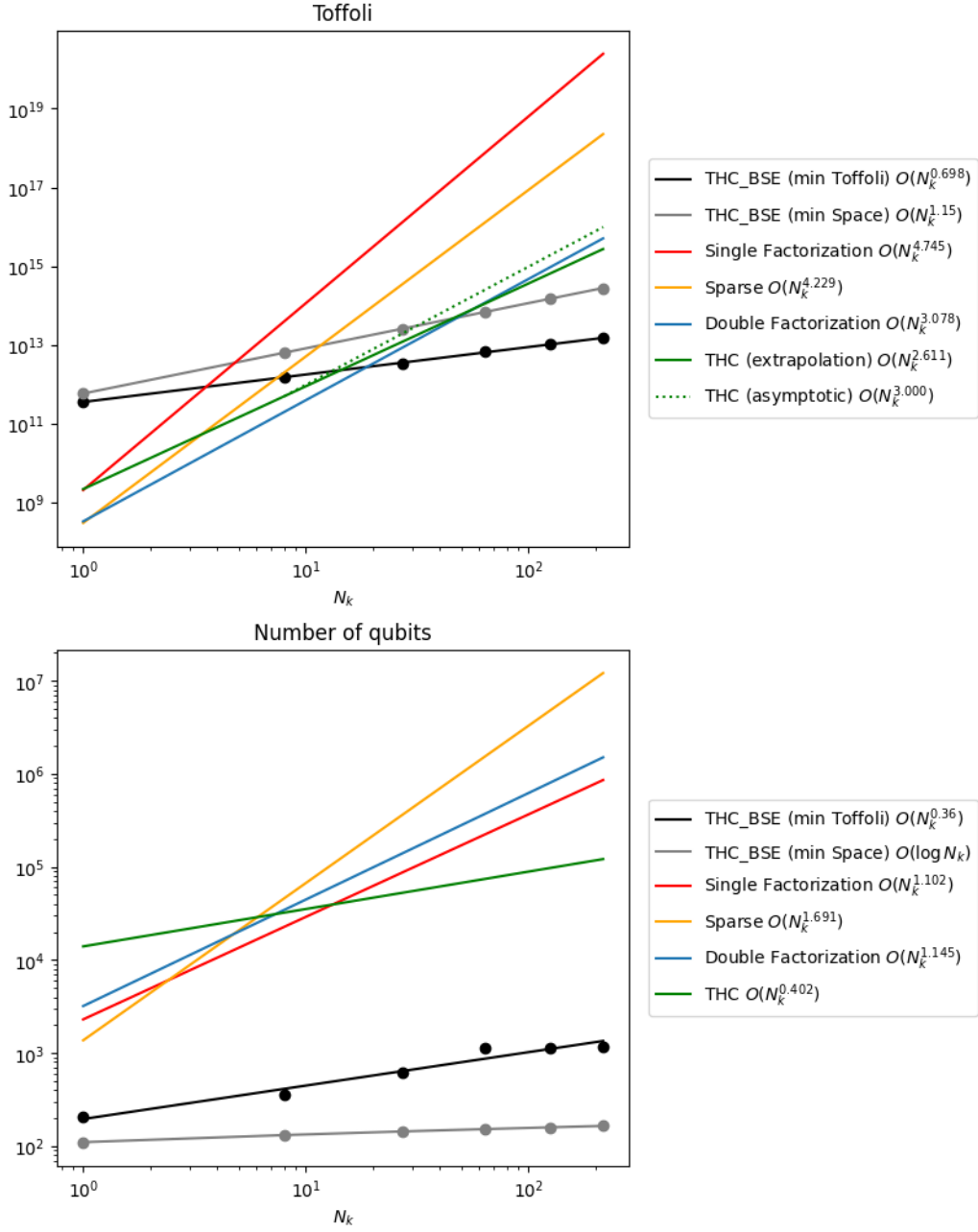
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where we get 188.

11 Comparison with other algorithms

Comparison with second-quantized symmetry-adapted algorithm [14].

Scheme	Total qubits	Toffolis per block-encoding	Total Toffolis	Empirical subnormalization λ (N_k only)	Empirical total Toffolis (N_k only)	Notes
Sparse	$\tilde{O}(N_k^{3/2}N^2)$	$\tilde{O}(N_k^{3/2}N^2)$	$\tilde{O}(N_k^{3/2}N^2\lambda_{\text{Sparse}}/\epsilon)$	$N_k^{2.578}$	$N_k^{4.229}$	
Single factorization	$\tilde{O}(N_kN^{3/2})$	$\tilde{O}(N_kN^{3/2})$	$\tilde{O}(N_kN^{3/2}\lambda_{\text{SF}}/\epsilon)$	$N_k^{2.857}$	$N_k^{4.745}$	
Double factorization	$\tilde{O}(\sqrt{N_k}N\sqrt{\Xi}) = \tilde{O}(N_kN^{3/2})$	$\tilde{O}(N_kN^{3/2})$	$\tilde{O}(N_kN^{3/2}\lambda_{\text{DF}}/\epsilon)$	$N_k^{2.289}$	$N_k^{3.078}$	Assumes average rank $\Xi = O(N_kN)$.
THC	$\tilde{O}(N_kN)$	$\tilde{O}(N_kN)$	$\tilde{O}(N_kN\lambda_{\text{THC}}/\epsilon)$	$N_k^{2.218}$	$N_k^{2.611}$	Observed sublinear scaling in N_k for Toffolis per block-encoding, but it could be a finite-size effect. Asymptotically, total Toffolis should be $\sim N_k^3$
BSE + THC (min Toffoli)	$\tilde{O}(\sqrt{N_k}N)$	$\tilde{O}(\sqrt{N_k}N^{3/2})$	$\tilde{O}(\sqrt{N_k}N^{3/2}\lambda_{\text{BSE}}/\epsilon)$	$N_k^{0.21}$	$N_k^{0.698}$	
BSE + THC (min Space)	$O(\log(N_kN))$	$\tilde{O}(N_kN^2)$	$\tilde{O}(N_kN^2\lambda_{\text{BSE}}/\epsilon)$	$N_k^{0.21}$	$N_k^{1.15}$	



Note: first quantization with Trotter actually has better scaling than Qubitization, see [Page 3 Eq 6](#)/[Page 14 Sec D.1](#)/[Appendix B](#)

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